

Pratham Arora

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EDUCATION

Plaksha University

B.Tech. in Computer Science & AI — CGPA: 8.19 / 10.0

Mohali, India

Expected Aug. 2026

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, C#, HTML/CSS, Bash
Backend & Web: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, GraphQL, Tailwind CSS
Databases: Azure Cosmos DB, PostgreSQL, Firebase, Vector Databases
Cloud & Tools: Azure, Git/GitHub, Linux, Vercel, RAG Pipelines

EXPERIENCE

AI Intern

June 2025 – July 2025

Cotality | Kolkata

- Collaborated with one other intern to build **FastAPI** backend processing 50,000+ lines of code daily across 15+ programming languages using Abstract Syntax Tree (AST) analysis, reducing engineering team's documentation time by an estimated 60%.
- Implemented hash-based change detection system for **vector database updates**, chunking codebases function-by-function and tracking content hashes to skip unchanged functions – reducing embedding generation costs by 85% by avoiding redundant processing.
- Developed 12 **RESTful API endpoints** with p95 latency under 200ms, designing response schemas to enable real-time code documentation preview in web interface.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs | Noida

- Refactored 15,000-line monolithic **RoboGPT** codebase into modular architecture with 40% fewer inter-module dependencies, enabling parallel development across 3 teams and reducing average bug fix time from 2 days to 6 hours.
- Integrated Computer Vision capabilities into robotic control loop, optimizing calibration routines and improving robotic arm positioning accuracy for complex food preparation tasks.

TECHNICAL PROJECTS

Resume Builder & Interview GPT | Next.js, TypeScript, Tailwind, Firebase, Gemini API

Jan. 2025 – April 2025

- Built full-stack AI resume builder serving 500+ users with real-time PDF preview, generating ATS-optimized resumes and achieving 4.2/5 average rating from 80+ user submissions.
- Integrated **Gemini 2.5 Flash API** to power "Interview GPT" feature generating 1,000+ personalized DSA questions with detailed solutions, reducing average user preparation time by 40% based on feedback surveys.
- Implemented **Firebase Authentication** with Google OAuth supporting 200+ active users, designed responsive UI with Tailwind CSS for mobile and desktop experiences.

Reddit Data Ingestion Pipeline | Python, PRAW, Firebase, Data Engineering

Aug. 2025 – Dec. 2025

- Built automated data pipeline using **PRAW** to scrape 22,000+ interactions from 19 pregnancy-related subreddits, implementing concurrent scraping with rate limiting and error recovery for 99.5% data completeness.
- Designed **Firebase NoSQL schema** with real-time deduplication logic using comment IDs and timestamps as composite keys, preventing duplicates across 5 parallel scraper instances.
- Implemented automated data cleaning and preprocessing scripts in Python, preparing unstructured text data for NLP analysis.

VR Conversational Agent | Unity, C#, Gemini 2.5 Flash, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud Speech-to-Text and Text-to-Speech, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.
- Designed low-latency interaction pipeline with asynchronous API orchestration and error handling for stable voice-based dialogue in immersive VR environment.
- Implemented embodied latency-masking mechanisms using procedural animations to maintain user immersion during network and inference delays.

LEADERSHIP

Head of Tech

April 2025 – Present

Athleda (Plaksha Sports Society) | Mohali

- Lead 4-person dev team building sports platform serving 500+ students with event registration, live scoring, and photo galleries – established CI/CD pipeline reducing deployment time from 45 minutes to 8 minutes.

Head of Web Development

Feb. 2024 – Feb. 2025

LEAP.AI@Plaksha | Mohali

- Managed 6-person web team building AI club website attracting 800+ monthly users – established team Git workflow reducing merge conflicts by 70% and mentored 6 junior developers in Next.js and TypeScript.